Neeraj KR

Software Engineer

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Portfolio: Neeraj K R

Professional Summary

Python Developer (B.Tech CSE, CGPA 9.15) with hands-on experience building scalable backend systems and RESTful APIs using Python, FastAPI, Django, Django REST Framework, with React frontends, PostgreSQL, AWS S3, and Docker. Experienced in AI-powered enterprise platforms, AI agent governance, and biometric device-integrated time & attendance software. Strong in database optimization, API design, authentication, role-based access control, debugging, and delivering reliable, production-ready features.

Skills

- **Languages:** Python (Proficient), SQL (Proficient), JavaScript (Decent), C programming, HTML5, CSS3
- **Backend:** Django, Django REST Framework (DRF), FastAPI, SQLAlchemy, Pydantic, Django ORM, MVT Architecture, Async/Await, Dependency Injection, Middleware
- **Frontend:** React.js, JavaScript (ES6+), HTML, CSS, DOM Manipulation, Django Templates, Responsive Design, Axios/Fetch API
- **AI / Data:** AI Chat (Natural-Language Queries), Retrieval-Augmented Generation (RAG), AI Agent Governance, Deep Learning, Image Processing, Genetic Algorithm
- **APIs & Security:** RESTful APIs, JSON, Serializers, ViewSets, Routers, Mixins, Pagination, Filtering, Authentication & Authorization, JWT, RBAC, Permissions, SHA-256 Hash Chain, Swagger/OpenAPI, Third-Party API Integration
- **Databases:** PostgreSQL, Oracle SQL*Plus, SQL Server, Joins, Aggregations, Query Optimization, Indexing, Model Relationships
- **Cloud & DevOps:** AWS S3, Docker, Docker Compose, Vercel, Git, GitHub, Linux Basics
- **Tools/Platforms:** VS Code, PyCharm, Postman, Swagger, pgAdmin 4, DBeaver, Jira, SourceTree, VMware, Antigravity, Cursor, IBM Bob
- **Core Concepts:** OOP, Data Structures, CRUD Operations, Exception Handling, File Handling, Debugging, Functional Testing, Notification Systems, Biometric Device Integration
- **Soft Skills:** Problem-Solving, Team Player, Communication, Adaptability, Quick Learner, Ownership

Experience

ZKTeco India — Bengaluru, Karnataka
Software Engineer Trainee — *Feb 2026 – Present*

- **AI-Powered Organization Management Platform**: Developing a centralized AI-powered web platform that helps organizations manage users, teams, files, permissions, notifications, and third-party integrations, with an AI chat feature for natural-language queries over organizational information.
  - Built scalable backend REST APIs using FastAPI with async endpoints, dependency injection, authentication, role-based permissions, and middleware.
  - Implemented user, team, and permission management modules, plus a notification system and third-party service integrations.
  - Implemented secure file upload, storage, and retrieval using AWS S3, with metadata and data managed in PostgreSQL via SQLAlchemy.
  - Developed responsive, interactive React interfaces and an AI chat experience for context-aware, natural-language assistance.
  - Tech: Python, FastAPI, React, PostgreSQL, SQLAlchemy, AWS S3, REST API, JWT, RBAC, Docker, Git.
- **Time & Attendance Management System**: Contributed to the development and maintenance of an enterprise time and attendance software integrated with biometric attendance devices.
  - Developed backend modules for attendance tracking, shift scheduling, leave management, and access control using Python, Django, and SQL.
  - Built and integrated REST APIs (DRF) for real-time software-to-device communication and data synchronization with biometric attendance devices.
  - Validated attendance data and optimized database queries and operations using SQL, improving performance and data accuracy.
  - Performed functional testing, debugging, and bug fixing to improve application reliability and stability.
  - Tech: Python, Django, Django REST Framework, SQL, REST API, HTML, CSS, JavaScript, Biometric Devices, Git.

Cloud Innovations
Full Stack with Python Intern — *Internship*

- Worked on full-stack web development using Python, Django, SQL, HTML, CSS, and JavaScript, building and testing application features.

Projects

- **Sentinel – AI Agent Trust & Governance Layer** (IBM Bob 2.0 Hackathon) GitHub
  Built a governance and audit layer for IBM Bob that intercepts agent actions, classifies risk with a rule-based policy engine, pauses high-risk steps for human approval, and records every event in a tamper-evident SHA-256 hash chain with on-demand tamper detection. Developed FastAPI services, a PostgreSQL schema, and a React dashboard with a live timeline, approval queue, and verification lab.
  Tech: Python, FastAPI, React (Vite), PostgreSQL, SQLAlchemy, Docker, SHA-256, Neon DB, Vercel
- **Brain Tumor Detection using Deep Learning and Genetic Algorithm:** Developed an automated brain tumor detection system for MRI images using image processing, deep learning, and Genetic Algorithm optimization, achieving **97%** accuracy.
  Tech: Python, Deep Learning, Image Processing, Genetic Algorithm
- **College Leave Management System:** Implemented a web-based system that automates leave applications, approvals, and tracking with role-based workflows.
  Tech: Python, Django, SQL, HTML, CSS, JavaScript
- **Portfolio Website:** Developed a responsive personal portfolio highlighting technical skills and projects.
  Tech: HTML, CSS, JavaScript

Education

Universal Engineering College Thrissur, Kerala
Bachelor of Technology – Computer Science and Engineering; CGPA: 9.15 *2021 – 2025*

Chentrappinni HSS Thrissur, Kerala
Higher Secondary Education; Percentage: 98.3% *2019 – 2021*

THSLC, Kodungallur Kerala
Secondary Education; Percentage: 98% *2019*

Languages

English, Malayalam